

Qatar-1b

R-1277

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_251007 · created 2026-07-31T14:34:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460955.682 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.029
Transit depth (Rp/Rs)^2	2.86 +/- 1.0 [%]
Orbital Inclination (inc)	82.57 +/- 1.82
Airmass coefficient 1 (a1)	0.952 +/- 0.021
Airmass coefficient 2 (a2)	0.038 +/- 0.037
Scatter in the residuals of the lightcurve fit is	2.24 %
Best Comparison Star	#2 - [299, 384]
Optimal Aperture	2.78
Optimal Annulus	10.96
Transit Duration (day)	0.058 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.029 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.058 d vs 0.0692 d (deviation 0.66σ)
Mid-time O–C	-3.5 min
Depth SNR	5.8
Residual scatter	2.24 %

Light curve

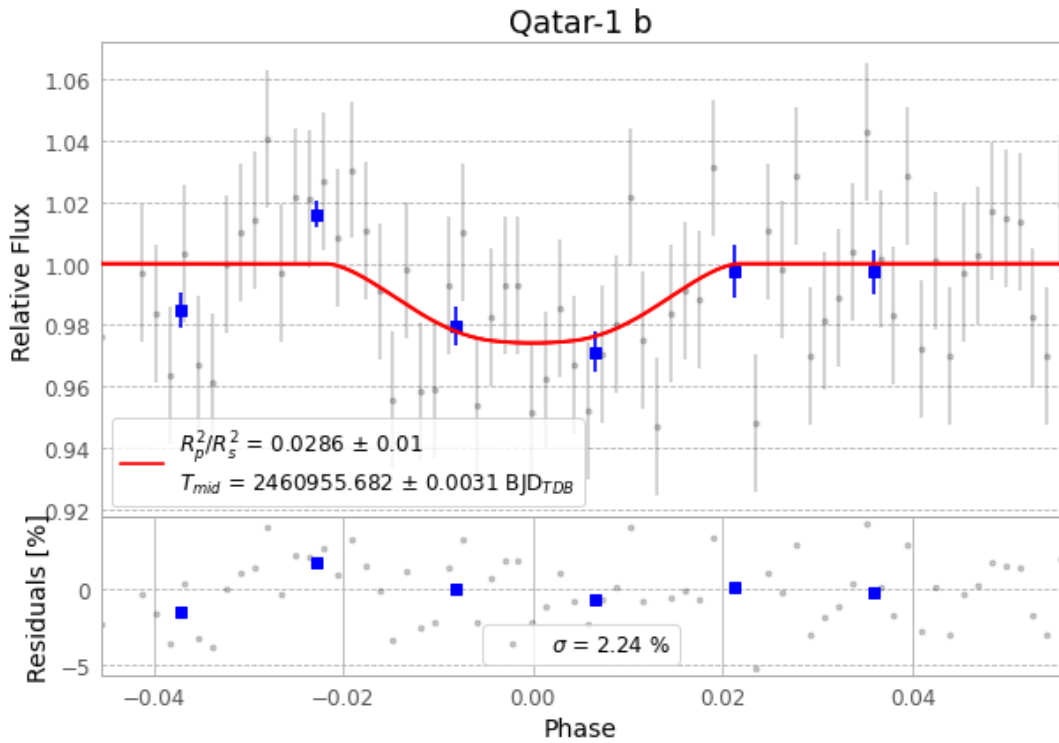


Figure 1 — FinalLightCurve_Qatar-1 b_2025-10-06.png (SHA-256 0124efcdf158a4ca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 362], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 328bb5ad3cbc8b49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), Qatar-1251007024518.FITS → Qatar-1251007061215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-251007.log (d7606c7e3059...), FinalLightCurve_Qatar-1 b_2025-10-06.png (0124efcdf158...), FinalLightCurve_Qatar-1 b_2025-10-06.csv (bd545f6b5773...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:34Z.